32S1 — Invariant Persistence, Relational Addressability, and Scale Transition

Professional reading edition of the evolving public MKUFT canon

Mark Charles McLaughlin

ORCID: 0009-0005-7736-1511

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1. Core refinement

A lower-scale degree of freedom need not be physically destroyed when a higher-scale organisation forms. The stronger and more testable possibility is that the lower-scale variable remains physically instantiated while losing **independent relational addressability** for the lower-scale individuality that previously possessed it.

The first formulation of this supplement treated the relevant identity class as though it could be held fixed while that loss was measured. That is not always lawful. If the system's boundary, maintenance relations, active construction address, or constitutive coupling to its environment changes strongly enough, the operational definition of the individuality under comparison may also change.

The corrected distinction is:

- **physical persistence** — the underlying variable or invariant remains realised;
- **independent addressability** — the current organised object can still vary, recruit, or route that variable as an independently available relation to itself;
- **individuality address** — the declared boundary, organisation, environment-coupling relations, history/lineage, and within-address equivalence rule under which states count as instances of that individuality;
- **address-class parity** — two temporally different individuality addresses may belong to the same declared address class even though their realised states or histories are not literally equal;
- **cross-address continuity** — when address-class parity fails, a separate test is required before the later individuality is called a continuation of the earlier one;
- **scale ownership** — a lower-scale variable may remain active while becoming bound into a higher-scale constraint relation rather than remaining a free coordinate of the earlier individuality.

> *The invariant can persist while its independent address changes.*

> *The self-address can also change. Do not assume the same self merely because the substrate persists.*

2. Operational individuality address and joint self-state space

Let the system state be

$$x_t \in \mathcal{X}_{\alpha_t},$$

where α_t is the active construction/state-space address inherited from Modules 31 and 32.

Let

$$g_t^{SE} \in \mathcal{G}_t^{SE}$$

denote the realised typed coupling state between candidate system S and relevant environment E. It may contain exchange, sensing, action, adhesion, boundary, regulatory, or other domain-appropriate relations. It is not a new physical layer.

Define the joint realised state

$$z_t = (x_t, g_t^{SE}) \in \mathcal{Z}_t, \qquad \mathcal{Z}_t = \mathcal{X}_{\alpha_t} \times \mathcal{G}_t^{SE}.$$

Let

$$\underset{t}{\overset{\text{state}}{\sim}}$$

be the declared **within-address state-equivalence rule** for the individuality claim at time t.

Define the **operational individuality address**

$$\sigma_t = \left(\alpha_t, \mathcal{B}_t, \mathcal{G}_t^{SE}, \mathcal{M}_t, H_t, \underset{t}{\overset{\text{state}}{\sim}}\right),$$

where $\mathcal{B}_t$ is the active boundary, $\mathcal{M}_t$ the relevant maintenance/closure relations, and H_t the retained lineage/history descriptor.

The corresponding operational self-state space is

$$\mathcal{S}_t = \mathcal{Z}_t / \underset{t}{\overset{\text{state}}{\sim}},$$

and the realised operational self-state is

$$s_t = [z_t]_{\overset{\text{state}}{\underset{t}{\sim}}} \in \mathcal{S}_t.$$

This is an operational systems object, not a metaphysical definition of personhood. The equivalence rule must be declared at the layer and scale of the actual claim.

2.1 Address-class parity

Exact tuple equality is generally too strong for evolving systems because realised history, boundary values, or coupling states may change lawfully while the criterion of individuality remains the same.

Define

$$\sigma_a \underset{\text{addr}}{\equiv} \sigma_b$$

when the two addresses belong to the same declared individuality-address class: differences between them are admissible under the same construction class, boundary class, coupling class, maintenance organisation, lineage rule, and within-address state-equivalence criterion.

This relation does **not** mean that the realised states are identical. It means that no constitutive criterion defining the address class has changed.

Thus three objects remain separate:

state equivalence ($\overset{\text{state}}{\underset{t}{\sim}}$), address-class parity ($\underset{\text{addr}}{\equiv}$), cross-address continuity ($\underset{\text{id}}{\approx}$).

3. Same-self parity guard

A scale transition can change not only state but the criterion under which states count as instances or continuations of an individuality.

If

$$\sigma_t \not\equiv_{\text{addr}} \sigma_{t+1},$$

because a constitutive boundary, environment-coupling class, maintenance relation, construction class, lineage criterion, or state-equivalence rule changed, then the statement

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"the same self lost degree of freedom i"
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is not automatically licensed.

A lawful cross-address continuation requires a declared map

$$T^{\text{id}}_{t\to t+1} : \mathcal{S}_t \supseteq \mathcal{D}_t \to \mathcal{S}_{t+1}.$$

Let

$$J_t : \mathcal{S}_t \to \mathcal{J}$$

be a declared identity-bearing invariant or invariant family. Cross-address continuity to tolerance ε_J requires

$$d_J\big(J_{t+1}(T^{\text{id}}_{t\to t+1} s_t), J_t(s_t)\big) \le \varepsilon_J,$$

plus any domain-specific lineage, viability, boundary, or maintenance conditions required by the identity claim.

Define

$$\sigma_a \underset{\text{id}}{\approx} \sigma_b$$

only when those cross-address continuation tests pass. This relation is not the same object as either $\overset{\text{state}}{\underset{t}{\sim}}$ or $\underset{\text{addr}}{\equiv}$.

If no lawful continuation map or identity-bearing invariant can be supplied, the correct description is not automatically `the same self changed`. The event may instead be a reconstituted individuality, incorporation into another individuality, replacement of the comparison class, or an unresolved identity transition.

> ***Same substrate is not same self. Same name is not same self. Same components are not same self. Prove the continuation relation at the address where identity is claimed.***

4. Joint reachable set and identity-relative reach

Module 32 defines reach on the addressed adaptive system. When a claim makes system–environment coupling constitutive, the reachable object must include the coupling state rather than silently projecting it away.

Let

$$\hat{\mathcal{R}}_t(z_t) \subseteq \mathcal{Z}_t$$

be the declared reachable set of joint system–environment realisations under the coupled dynamics used by the implementation.

Let

$$\mathcal{V}_{\sigma_t} \subseteq \mathcal{Z}_t$$

be the joint realisation set compatible with persistence of the individuality class specified by σ_t. Define

$$\mathcal{R}^{\mathrm{id}}_{\sigma_t}(z_t) = \hat{\mathcal{R}}_t(z_t) \cap \mathcal{V}_{\sigma_t}.$$

This contains joint states reachable without leaving the declared individuality class under the active boundary, coupling, maintenance and viability conditions.

A lower-scale internal variable or a system–environment relation may therefore remain physically present while becoming impossible to vary independently inside this identity-relative set.

If the implementation has no justified coupled dynamics for g^{SE}, it may fall back to Module 32's system-only reachable set, but it must then **not** make quantitative claims about independent addressability of environment-coupling relations.

If the before/after addresses fail address-class parity, the old and new reachable sets must not be compared as though one fixed individuality class had silently persisted.

5. Independent relational addressability

Let

$$I_i : \mathcal{Z}_t \to \mathcal{Y}_i$$

be a declared internal, boundary, or system–environment relation-coordinate, and let

$$\mathcal{A}_\ell(t) = \{I_1, I_2, \ldots, I_n\}$$

be the family independently addressable before consolidation.

For tolerance pair $(\varepsilon_i, \delta_\sigma)$, call I_i **independently addressable under individuality address $\boldsymbol{\sigma_t}$** when there exist

$$z', z'' \in \mathcal{R}^{\mathrm{id}}_{\sigma_t}(z_t)$$

such that

$$||I_i(z') - I_i(z'')|| > \varepsilon_i,$$

while the other declared identity-bearing observables required to remain fixed for the test vary by no more than δ_σ.

Define

$$\mathcal{A}_{\ell|\sigma_t} = \{I_i \in \mathcal{A}_\ell(t) : I_i \text{ passes the identity-relative addressability test}\}.$$

This concerns **which relations remain independently deployable while remaining within the declared individuality class**. It does not say that the underlying molecules, forces, variables, or physical possibilities vanished.

6. Comparison of addressability across time or address

If

$$\sigma_{t_1} \underset{\text{addr}}{\equiv} \sigma_{t_2},$$

ordinary before/after comparison may be available under a declared common observable or measure μ_A.

If address-class parity fails but cross-address continuity is established,

$$\sigma_{t_1} \not\equiv_{\text{addr}} \sigma_{t_2}, \qquad \sigma_{t_1} \underset{\text{id}}{\approx} \sigma_{t_2},$$

then an additional map is required for the particular addressability relation being compared. Let

$$T_A : \mathcal{A}_{\ell|\sigma_{t_1}} \to \tilde{\mathcal{A}}_{\ell|\sigma_{t_2}}$$

be that lawful comparison map. Define

$$\Delta_{T_A} A_\ell = \mu_A\Big(\mathcal{A}_{\ell|\sigma_{t_2}}\Big) - \mu_A\Big(T_A(\mathcal{A}_{\ell|\sigma_{t_1}})\Big).$$

If no justified comparison map exists, report the two addressability structures separately rather than manufacturing a scalar difference.

This is the **same-self parity extension** of Module 32's no-false-subtraction rule.

7. Constraint binding rather than destruction

Suppose a lower-scale action or movement set before consolidation is

$$\mathcal{U}_\ell^{\text{self}}(z).$$

Under a higher-scale organisation with individuality address σ_L, the identity-compatible subset may be

$$\mathcal{U}_{\ell|\sigma_L}^{\text{self}}(z) = \mathcal{U}_\ell^{\text{self}}(z) \cap \mathcal{U}_{\sigma_L}^{\text{adm}}(z),$$

with

$$\mathcal{U}_{\ell|\sigma_L}^{\text{self}}(z) \subsetneq \mathcal{U}_\ell^{\text{self}}(z).$$

The excluded routes are not necessarily physically impossible in isolation. They are inaccessible **while remaining within the organised identity class being tested**.

If address-class parity fails across the transition, the same-self parity guard must be satisfied before the phrase `the old self lost access` is used.

8. Three scale-transition identity classes

8.1 Same-address-class restriction

The individuality address evolves within the same declared address class and lower-scale independent addressability decreases:

$$\sigma_{t_1} \underset{\text{addr}}{\equiv} \sigma_{t_2}, \qquad \Delta A_\ell < 0.$$

Here it is meaningful to say that the same operational individuality class became more constrained, subject to the declared tolerance and comparison observable.

8.2 Readdressed continuation

The individuality address changes class but passes the separate continuity test:

$$\sigma_{t_1} \not\equiv_{\text{addr}} \sigma_{t_2}, \qquad \sigma_{t_1} \underset{\text{id}}{\approx} \sigma_{t_2}.$$

Here the individuality may be said to persist through reorganisation, but comparisons of freedom, capability, state, or self require the relevant cross-address maps.

8.3 Identity replacement, incorporation, or unresolved transition

The individuality address changes class and cross-address continuity has not been established:

$$\sigma_{t_1} \not\equiv_{\mathrm{addr}} \sigma_{t_2}, \qquad \sigma_{t_1} \not\approx_{\mathrm{id}} \sigma_{t_2} \quad \text{or continuity is unresolved.}$$

The formalism therefore does **not** call the pre- and post-transition objects the same self by default. A lower-scale individuality may have been incorporated into a higher-scale whole, replaced as the relevant comparison object, or left unresolved with respect to identity continuity.

These cases must not be collapsed into one word such as `emergence`.

9. Scale-transition onset and completion

A scale transition should not be defined only by the later appearance of a new capability.

Where address-class parity or readdressed continuity has been established, a provisional onset may be associated with sustained reduction in lower-scale independent addressability under an emerging load-bearing closure:

$$t_{\mathrm{access}} = \inf\Big\{t : \Delta_{T_A} A_\ell(t) < -\eta_A \text{ and } \Gamma_L(t) \geq \tau_\Gamma\Big\},$$

where Γ_L is a declared measure of higher-scale closure or organisational persistence. Within one address class, T_A may reduce to the declared common-coordinate comparison.

A fuller transition requires separately measured higher-scale gain. Schematically,

$$\Delta_{T_A} A_\ell < 0, \qquad \Delta_{T_L} A_L > 0, \qquad \Delta_{T_K} K_L > 0,$$

where each T is the lawful comparison map for its own observable; no universal transport map is assumed.

If the individuality criterion changes class and no continuation map is available, the earliest defensible marker is instead the **identity-address transition**: the point at which the old comparison class ceases to support the observed organisation and a new one becomes necessary.

This separates three events that may coincide or occur in sequence:
- **access loss** — formerly independent relations become bound;
- **identity readdressing** — the operational criterion of self-continuation changes;
- **higher-scale gain** — new stable relations or capabilities become available.

10. Environment coupling can be constitutive

Environment is not automatically constitutive of identity, but it cannot be assumed incidental either.

Module 31 already carries environment $\mathcal{E}_t$, boundary $\mathcal{B}_t$, history and support conditions inside the context descriptor because changing them can alter object class or admissible dynamics.

For an individuality claim, test the system–environment coupling directly. Let D_E be a controlled deformation of the environment or system–environment relation that preserves as much internal component state as the domain permits.

If

$$D_E[\sigma_t] \notin \mathcal{M}_{\mathrm{id}},$$

where $\mathcal{M}_{\mathrm{id}}$ is the declared same-individuality comparison class, then the changed coupling is constitutive for that identity claim rather than merely background context.

This does not mean the environment `is the self`. It means some relations to the environment may participate in defining which organised individuality is actually present.

11. Distinguishing endogenous address change from observer coarse-graining

Ordinary coarse-graining can remove fine detail from **our model** while leaving the real system unchanged. A claimed scale or self-address transition therefore needs more than successful macroscopic compression.

Useful controls are: observer-only coarse-graining; release of the proposed higher-scale constraint while preserving lower machinery; controlled deformation of candidate constitutive system–environment coupling; boundary-shift tests; preregistered identity-map falsification; component-preservation tests; and a separately measured higher-scale-gain control.

The decisive question is whether the real system's available relations change, not merely whether our description uses fewer coordinates.

12. Example geometry — motile cells to organised collective

For a motile lower-scale unit, a free heading or movement repertoire can remain physically supported by cytoskeleton, motors, receptors and energy supply while adhesion, signalling, polarity, collective geometry and environmental coupling restrict which trajectories remain independently available.

A serious test should establish that the machinery for the old trajectory class remains present; release from the organising relation restores part of the old repertoire where reversibility is predicted; restriction occurs selectively while higher-scale coordinated movement improves; candidate constitutive environment coupling is measured rather than assumed; a new collective coordinate becomes controllable or predictively useful; and any same-self claim across changed individuality-address classes supplies a continuation map.

A strong incorporation pattern is therefore not merely

$$\Delta A_\ell < 0, \qquad \Delta K_L > 0,$$

but the stronger conjunction

$$\text{lower machinery persists} \wedge \text{identity relation is declared} \wedge \Delta_{T_A} A_\ell < 0 \wedge \Delta_{T_K} K_L > 0.$$

This supports **constraint incorporation under an explicit individuality address**, rather than literal destruction of the lower variables.

13. Relationship to established neighbours

This refinement sits near substantial prior work and does not claim the broad ingredients as inventions.

Varela, Maturana and Uribe (1974) formulate autopoietic organisation as the organisation that makes a living system an autonomous unity. Di Paolo (2005) develops adaptivity and sense-making as regulation relative to conditions of viability. Moreno and Etxeberria (2005) treat self-construction and activity in the environment as aspects of one organisation in basic autonomous systems. Barandiaran, Di Paolo and Rohde (2009) explicitly require an agent to define its individuality and describe agency as autonomous organisation adaptively regulating its coupling with the environment. Aguilera and Di Paolo (2018; 2019) use perturbational/information-integration methods to delimit an integrated unit from its environment and model adaptive preservation of integrity under environmental change. Coarse-graining, quotient spaces, multiscale modelling, controlled/uncontrolled manifolds and organisational closure also supply established neighbouring tools.

The broad claim that individuality is relational, self-maintaining, boundary-dependent, or coupled to environment is therefore **not** attributed to MKUFT as a first discovery.

The present MKUFT refinement is narrower: **apply addressed-state/no-false-parity discipline to the individuality criterion itself, so that changing constitutive boundary or environment coupling can readdress the self-state and make naive before/after freedom comparisons undefined until a lawful identity-continuation relation and observable-specific comparison maps are supplied.**

Historical priority for that exact integration is not asserted here.

Relevant references include:

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- Aguilera, M. and Di Paolo, E. A. (2019). *Integrated information in the thermodynamic limit.* Neural Networks 114, 136–146. DOI 10.1016/j.neunet.2019.03.001.
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14. Failure and reduction conditions

This refinement is weakened or reduced if the lower capability disappears because its physical machinery is destroyed rather than relationally bound; apparent access loss exists only in the observer's coarse-grained model; changing environment coupling fails to alter the claimed individuality class where predicted; the same-self continuation map is post-hoc or fails preregistered invariants; releasing the proposed organising constraint fails to restore lower autonomy where reversibility is predicted; no stable higher-scale closure/capability appears; the higher-scale object adds no predictive/interventional value; or the result depends on arbitrary post-hoc identity rules, thresholds, macrostate definitions or environment boundaries.

If those failures occur, use the simpler physical, organismic, dynamical or coarse-grained description instead.

15. Claim boundary

This module does **not** establish that every emergence event is a scale transition of this form. It does not establish a universal sharp transition time, consciousness, survival after death, reincarnation, `source`, or independent I⟶P dynamics.

It also does not claim that identity is always environmentally extended. Environment coupling must earn constitutive status for the specific individuality claim under test.

The scientifically useful claim is narrower:

> ***A candidate scale transition can require simultaneous auditing of physical persistence, independent relational accessibility, and identity parity. Lower variables can persist while higher-scale closure removes their independent deployment; if constitutive system–environment relations or the within-address identity rule also change class, the self-state is readdressed and same-self comparison requires a separate lawful continuation relation.***

16. Compressed rules

> ***Do not ask only whether the old degree of freedom still exists. Ask whether the old individuality can still address it independently while remaining within a lawfully comparable identity class.***

> ***If the relation to the environment changes the criterion of individuality, prove same-self parity before comparing old and new freedom.***

> ***State equivalence, address-class parity, and cross-address self continuity are three different relations. Never let one silently stand in for another.***

> ***The invariant may persist while its address changes. The self-address may change with it.***

> ***A scale transition can begin with load-bearing loss of lower-scale addressability, but higher-scale completion and identity continuity are separate questions that must each earn their own comparison map.***